

ROYAL FITNESS CLUB

RFC PREMIUM GUIDE #15

RECOVERY, SLEEP & CNS RESTORATION

Complete System for Maximising Recovery Between Sessions

5 Chapters · Sleep Science · CNS Monitoring · Evidence-Based

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DISCLAIMER

This guide is for educational purposes only. Not medical advice. Consult a qualified healthcare professional before starting any protocol. Individual results vary.

CHAPTER 1 — RECOVERY: YOUR MOST IMPORTANT VARIABLE

"You do not get fit from training. You get fit from recovering from training. Training is only the stimulus."

Recovery is the biological process by which the body repairs and adapts to the stress of training. Muscle protein synthesis, glycogen replenishment, hormonal restoration, and connective tissue repair all occur during recovery — not during training. The training session provides the stimulus; recovery provides the adaptation. Elite athletes do not train harder than amateurs — they recover better.

Recovery System	Primary Process	Time Required	Bottleneck Factor
Muscle tissue	Protein synthesis, satellite cell activation	24–72 hours	Protein intake, sleep quality
Glycogen	Glucose conversion and storage	12–24 hours	Carbohydrate intake timing
Central Nervous System	Neurotransmitter replenishment	24–96 hours (heavy sessions)	Sleep, intensity management
Connective tissue	Collagen synthesis, tendon remodelling	72 hours to weeks	Vitamin C, collagen, hydration
Hormonal	Cortisol normalisation, T restoration	24–48 hours	Sleep, stress management

CHAPTER 2 — SLEEP ARCHITECTURE & OPTIMISATION

*"The majority of your testosterone and growth hormone is released during deep sleep.
Miss the sleep; miss the gains."*

Sleep Stage	Duration per Cycle	Recovery Function	Disrupted By
N1 (Light Sleep)	5–10 min	Transition to deeper stages	Noise, light, anxiety
N2 (NREM)	20–25 min	Memory consolidation, cell repair	Inconsistent schedule
N3 (Deep / SWS)	20–40 min	GH release, muscle repair, immune function	Alcohol, stimulants, stress
REM	20–25 min	Brain recovery, emotional processing	Alcohol, THC, late meals

Sleep Optimisation Protocol

- Consistent sleep/wake schedule — same time 7 days per week (most important variable)
- Room temperature: 18–20°C — core temperature drop initiates and maintains sleep
- Complete darkness: blackout curtains or sleep mask — even faint light disrupts melatonin
- No screens 60–90 min before bed — blue light suppresses melatonin by up to 50%
- No alcohol: disrupts REM and N3 sleep even at low doses
- Last meal 2–3 hours before bed — digestion increases core temperature
- Magnesium glycinate 300mg before bed — proven to increase deep sleep duration
- Consistent morning light exposure: 10+ min direct sunlight anchors circadian rhythm

CHAPTER 3 — CNS RECOVERY MONITORING

The Central Nervous System (CNS) is often the limiting factor in high-frequency training. Unlike muscles, CNS fatigue is not visible and is easily confused with laziness or lack of motivation. Systematic monitoring prevents overtraining and ensures training quality remains high.

CNS Fatigue Indicator	How to Measure	Action Threshold	Recovery Protocol
Morning resting HR	Check on waking before rising	+7 BPM above baseline	Reduce intensity/volume that day
Grip strength test	Dynamometer or hang test	> 5% below baseline	De-load or rest day
HRV (Heart Rate Variability)	HRV4Training app, Oura Ring	> 2 RMSSD below 7-day avg	Active recovery only
Mood/motivation (1–10)	Daily journal rating	Score < 5 for 3+ days	Mandatory rest or deload week
Sleep quality rating	Morning score 1–10	< 6 for 3+ days	Address sleep environment first

■ **A deload week (50–60% of normal volume at normal intensity) every 4–6 weeks prevents accumulated CNS fatigue. Plan it proactively — do not wait for forced rest from injury.**

CHAPTER 4 — ACTIVE RECOVERY TECHNIQUES

Technique	Mechanism	Duration	Frequency	Evidence Grade
Zone 2 Walking	Blood flow, lactate clearance, fat oxidation	30–45 min	Daily	A (very strong)
Foam Rolling	Fascial pressure, pain reduction, mobility	10–15 min	Post-training	B (good)
Contrast Showers	Vasoconstriction/dilation, lymphatic	10 min (alternating)	4× per week	B (good)
Yoga / Stretching	ROM, parasympathetic activation	20–30 min	3–4× per week	B (good)
Cold Water Immersion	Inflammation reduction (acute)	10–15 min at 10–15°C	2–3× per week	B (moderate)
Sauna	Heat shock proteins, GH release	15–20 min at 80–100°C	2–3× per week	B (growing)
Massage	DOMS reduction, tissue quality	60 min	Weekly	A (strong for DOMS)

CHAPTER 5 — SUPPLEMENT STACK FOR RECOVERY

"The best recovery supplement stack contains nothing exotic — sleep, protein, creatine, and consistency outperform every trending product."

Supplement	Recovery Role	Dose	Timing	Priority
Magnesium Glycinate	Deep sleep quality, muscle relaxation	300–400mg	Before bed	Essential
Creatine	PCr replenishment, cell hydration	5g/day	Any time (consistent)	Essential
Whey / Casein Protein	MPS substrate provision	30–40g	Post-workout + pre-sleep	Essential
Tart Cherry Extract	Antioxidant, DOMS reduction, melatonin	480mg twice daily	Morning + evening	High
Collagen + Vitamin C	Tendon/ligament synthesis	15g + 500mg Vit C	1h pre-training	High
Ashwagandha	Cortisol reduction, sleep quality	600mg	Evening	Moderate
Melatonin (if needed)	Sleep onset (circadian support)	0.5–1mg	30 min before bed	Situational
BPC-157 (advanced)	Accelerated tissue repair	250–500mcg/day	SubQ near injury	Advanced only

- Recovery nutrition window: 30g protein + 50–80g carbs within 2 hours post-training
- Hydration: minimum 3L/day — dehydration by even 2% significantly impairs recovery
- Alcohol: even 2–3 drinks reduces muscle protein synthesis by 37% for 12–24 hours